Exam 1 Recap (review slides)

Module 1 - Introduction & Math Review · Review set · 9 Sep 2026

Name _____ Date _____

A solution of a drug was freshly prepared at a concentration of 300 mg/mL. After 30 days, the drug concentration in the solution was 80 mg/mL.

Module 1 practice, item 1

Assuming first-order kinetics, when will the drug decline to one-half of the original concentration?

Module 1 practice, item 2

Assuming zero-order kinetics, when will the drug decline to one-half of the original concentration?

Module 3 practice, item 1

Recommend an infusion rate to achieve a steady-state concentration of 24 mg/L of an agent that has an elimination half-life of 5 hours and a volume of distribution of approximately 22 L.

Module 3 practice, item 2

Recommend a loading dose that will achieve a steady-state concentration of 24 mg/L for an agent whose volume of distribution is approximately 22 L.

Module 3 practice, item 3

The same agent, with an elimination half-life of 5 hours and a volume of distribution of 22 L, is infused at the rate that gives a steady-state concentration of 24 mg/L. What is the expected plasma concentration 10 hours after the start of the infusion?

Module 3 practice, item 4

An agent with an elimination half-life of 5 hours and a volume of distribution of 22 L is infused at the rate that would give a steady-state concentration of 24 mg/L. The infusion is stopped 5 hours after it was started. What is the plasma concentration 3 hours after cessation?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

Module 1 practice, item 1 Worked by the drill

Answer: 15.7 days

WHAT THE STEM GIVES

C_0	300 mg/mL	the numerator inside ln(C_0 over C)
t	30 days	the denominator of k = ln(300 over 80) / 30
C	80 mg/mL	the denominator inside the logarithm

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(C_0 \div C)}{t} = \frac{\ln(300 \div 80)}{30} \text{ days.}$$

Two concentrations and the time between them determine the first-order rate constant, and here the earlier concentration is the initial one, so no back-extrapolation is needed. The milligram per millilitre units cancel inside the logarithm. The elapsed time is in days, so the rate constant will be per day.

ALGEBRA

$$k = \frac{1.32176}{30} \text{ days} = 0.04406 \text{ day}^{-1}, \text{ which Dr. Mosley reports as } 0.044 \text{ day}^{-1}.$$

A pure number divided by days gives reciprocal days. The elapsed time is 30 days, so k is per day, not per hour.

SETUP

$$t½ = \frac{0.693}{k}.$$

The question asks when the concentration reaches one half of its original value, which is the definition of the half-life. Dr. Mosley notes that the wording, declines to one-half, is what identifies a half-life question.

ALGEBRA

$$t½ = \frac{0.693}{0.04406} \text{ day}^{-1} = 15.73 \text{ days.}$$

Dividing a dimensionless logarithm by a quantity in reciprocal days leaves days, so the result is a time and carries no reciprocal unit. As a check on direction, the concentration has already fallen well past half of 300 by day 30, so a half-life shorter than 30 days is what the data demand.

ROUND

$$t½ = 15.7 \text{ days.}$$

Her worked answer prints 15.7 days, to one decimal place, which matches the precision of a rate constant she reports as 0.044 per day. Carrying the unrounded rate constant instead gives 15.73 days, a difference far too small to change any conclusion drawn from it.

DOES THE NUMBER MAKE SENSE?

300 falls to 80 mg/mL in 30 days, a factor a little under 4, so 30 days is a little under two half-lives and t½ is a little over 15 days; 15.7 days fits.

Module 1 practice, item 2 Worked by the drill

Answer: 20.45 days

WHAT THE STEM GIVES

C_0	300 mg/mL	the drop $C_0 - C = 220$ mg/mL, then the numerator of $\frac{C_0}{2k}$
t	30 days	the denominator of $k = \frac{220}{30}$
C	80 mg/mL	subtracted from C_0 for the drop

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{C_0 - C}{t} = (300 - 80)\ \text{mg/mL} \div 30\ \text{days}.$$

Under the zero-order assumption the concentration falls linearly, so the rate constant is the quantity lost divided by the time taken. The same two data points are used as in the first-order version; only the relation applied to them changes. Milligrams per millilitre divided by days gives milligrams per millilitre per day.

ALGEBRA

$$k = \frac{220}{30} = 7.333\ \text{(mg/mL)/day, which Dr. Mosley reports as } 7.33\ \text{(mg/mL)/day}.$$

The rate constant is a rate here rather than a proportion, so it carries concentration per unit time. Every day the solution loses the same 7.33 mg/mL whatever remains, which is the assumption being tested.

SETUP

$$t_{½} = \frac{C_0}{2k}.$$

Half of 300 mg/mL has to be removed, and the rate constant states how much is removed per day, so the time is the quantity to remove divided by that rate. Unlike the first-order case, the starting concentration appears explicitly and the answer would change if it did.

ALGEBRA

$$t_{½} = 300\ \text{mg/mL} \div [2 \times 7.333\ \text{(mg/mL)/day}] = \frac{300}{14.666} = 20.45\ \text{days}.$$

The concentration units cancel and the reciprocal day inverts, leaving days. The result is a time, worked here from the unrounded rate constant of 7.333 (mg/mL)/day.

ROUND

$$t_{½} = 20.45\ \text{days}.$$

Dr. Mosley reports 20.45 days, to two decimal places, worked from the unrounded rate constant. The zero-order half-life is far longer than the first-order one for the same pair of points, because a zero-order process removes a fixed amount per day rather than a fixed fraction.

DOES THE NUMBER MAKE SENSE?

> Losing 220 mg/mL took 30 days and half of 300 mg/mL is less than 220, so the half-life is under 30 days; at 7.33 (mg/mL)/day, 300 over 14.666 gives 20.45 days.

Rounding k to 7.33 before dividing gives 20.46 days; carrying 7.3333 gives 20.45 days. Both are accepted.

Module 3 practice, item 1 Worked by the drill

Answer: 73.18 mg/hr

WHAT THE STEM GIVES

C_{ss}	24 mg/L	the target, multiplied by Cl
t½	5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	22 L	with k gives Cl = 0.1386 × 22 L

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = C_{ss} \cdot Cl = C_{ss} \cdot k \cdot V_D$$

The rate that holds a chosen plateau is that plateau multiplied by clearance, because at steady state the amount delivered per hour has to match the amount removed per hour. Writing clearance as k times V_D lets the half-life and the volume given in the stem be used directly.

ALGEBRA

$$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The rate constant is what appears in the clearance, so the half-life is converted first. Dividing 0.693 by a time in hours leaves reciprocal hours.

ALGEBRA

$$R = (24\ mg/L)(0.1386\ hr^{-1})(22\ L) = 73.18\ mg/hr$$

The litres cancel between the volume of distribution and the concentration, leaving milligrams per hour. The clearance implied here is 3.049 L/hr, and 24 multiplied by that is 73.18.

ROUND

R = 73.18 mg/hr, which she rounds to 73 mg/hr as the practical rate

Two decimal places suit the four-figure rate constant, and 73.18 is the computed value. Dr. Mosley rounds infusion rates to a whole number in her recommendations, so 73.18 and her 73 mg/hr are both accepted; the unit is mg per hour because R is an amount per unit time.

> **DOES THE NUMBER MAKE SENSE?**
> k is 0.693 over 5, 0.1386 hr^{-1}, so Cl is 0.1386 × 22, about 3 L/hr; 24 mg/L × 3 L/hr is about 24 × 3 mg/hr, and 73.18 fits.

Module 3 practice, item 2 Worked by the drill

Answer: 528 mg

WHAT THE STEM GIVES

C_{ss}	24 mg/L	the target concentration, one factor

V_D	22 L	the volume to fill, the other factor

THE WORKING, ONE LINE AT A TIME

SETUP

$D_L = C_{ss} \cdot V_D$

The loading dose must put into the body the amount of drug that would be there at steady state, and volume of distribution is the constant that converts a plasma concentration into an amount in the body. No rate constant is needed by this route, which is why the half-life does not appear.

ALGEBRA

$D_L = (24\ \text{mg/L})(22\ \text{L}) = 528\ \text{mg}$

Milligrams per litre multiplied by litres cancels the litres and leaves milligrams, which is what a dose must be. The arithmetic is 24 times 22.

ROUND

$D_L = 528\ \text{mg}$

She states 528 mg in the review session and reports it as a whole number of milligrams, as she does for every loading dose in this module.

> **DOES THE NUMBER MAKE SENSE?**
> A loading dose fills the volume to the target at once: 24 mg/L × 22 L = 528 mg, in mg because the litres cancel.

Module 3 practice, item 3

Worked by the drill

Answer: 18 mg/L

WHAT THE STEM GIVES

$t_{½}$	5 hours	10 hr is two half-lives, which fixes the fraction 0.75
V_D	22 L	*not needed: the plateau is given, so no clearance enters*
C_{ss}	24 mg/L	the plateau multiplied by the fraction reached
t	10 hours	two half-lives of 5 hours

THE WORKING, ONE LINE AT A TIME

SETUP

$10\ \text{hr} \div 5\ \text{hr} = 2\ \text{half-lives}$

The stated time is a whole number of half-lives, which she points out is deliberate. Once the elapsed time is expressed in half-lives the fraction of steady state achieved can be read off the table rather than computed.

ALGEBRA

Fraction of C_{ss} at 2 half-lives = 0.75

Each half-life closes half of the remaining gap, so after two half-lives one quarter of the gap remains and three quarters has been covered. This holds for every drug, because the fraction depends only on the number of half-lives.

ALGEBRA

$C_p = (0.75)(24\ \text{mg/L}) = 18\ \text{mg/L}$

Multiplying the plateau by the fraction achieved gives the concentration at that time. The fraction is dimensionless, so the answer keeps the units of the steady-state concentration.

ALGEBRA

Long route: $C_p = (24\ \text{mg/L})(1 - e^{-(0.693/5\ \text{hr})(10\ \text{hr})}) = 18\ \text{mg/L}$

Evaluating the exponential directly gives a bracket of 0.75 and the same 18 mg/L, which is the check she prints beside the half-life reasoning. Both routes are accepted, and she reports 18 mg/L.

ROUND

$C_p = 18\ \text{mg/L}$

Both the half-life reasoning and the exponential give exactly 18 mg/L, so nothing is rounded away and the whole number stands. She reports 18 mg/L, and the agreement of the two routes is the check she prints beside the short one.

> **DOES THE NUMBER MAKE SENSE?**
> Ten hours is two half-lives of 5 hours, so the infusion has climbed half, then half of the rest: 50 → 75 per cent of 24 mg/L, 18 mg/L; the level cannot exceed 24.

Module 3 practice, item 4

Worked by the drill

Answer: 7.92 mg/L

WHAT THE STEM GIVES

t½	5 hours	5 hr of infusion is one half-life; also $k = \frac{0.693}{5} = 0.1386\ \text{hr}^{-1}$
V_D	22 L	*not needed: the plateau is given*
C_{ss}	24 mg/L	the plateau; half of it, 12 mg/L, is C_{peak}
infusion time	5 hours	one half-life, so the fraction reached is 0.5
t	3 hours	the time since cessation: $kt = 0.1386 \times 3$

THE WORKING, ONE LINE AT A TIME

SETUP

5 hr of infusion = 1 half-life, so the fraction of C_{ss} achieved is 0.5

The infusion was stopped well before steady state, so the plateau cannot be used as the starting concentration. One half-life of infusion closes half the gap, so the concentration at cessation is half of the steady-state value.

ALGEBRA

$C_{peak} = (0.5)(24\ \text{mg/L}) = 12\ \text{mg/L}$

This is the concentration at the moment the infusion stops, and it is what the decay begins from. She corrects a student who answers 18 mg/L here, which would be the two-half-life figure.

SETUP

$C_p = C_{peak} \cdot e^{-kt}$, with t counted from the moment the infusion stopped

After cessation nothing is going in, so the concentration follows ordinary first-order elimination from the peak, exactly as it would after a bolus. The 3 hours is counted from cessation and not from the start of the infusion, which is the single most common place to lose this question.

ALGEBRA

$$k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1}$$

The decay is written in terms of the rate constant, so the stated half-life has to be converted before the exponential can be evaluated. Dividing 0.693 by a time in hours leaves reciprocal hours, which is what a first-order rate constant must carry.

ALGEBRA

$$C_p = (12\ \text{mg/L})e^{-(0.1386\ \text{hr}^{-1})(3\ \text{hr})} = (12\ \text{mg/L})(0.6598) = 7.9170\ \text{mg/L}$$

The exponent is dimensionless because reciprocal hours multiply hours, and the concentration keeps its own units. Three hours is a little over half a half-life, so the answer should sit between 12 and 8.5 mg/L, which it does.

ROUND

$C_p = 7.92\ \text{mg/L}$

Two decimal places follows the precision carried through the rest of this problem set. 7.9 mg/L, the same figure to one decimal place, is also accepted on her paper.

DOES THE NUMBER MAKE SENSE?

Five hours of infusion is one half-life, so the level reached half of 24 mg/L, 12 mg/L; 3 hours after stopping is less than a half-life, so between 6 and 12 mg/L remains, and 7.92 fits.